

Program: Diploma in Renewable Energy	
Course Code: 4292	Course Title: DC Machines & Transformers
Semester: 4	Credits: 3
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To empower students to understand the working of electrical equipment used in everyday life and to expose the students to the concepts of various types of electrical machines and applications of electrical machines.
- To analyze power requirements, power capability, efficiency, operating characteristics, control requirements and electrical demands of these machines.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic Electrical Engineering		Fundamentals of Electrical & Electronics Engineering	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the classification and characteristics of DC generators	15	Understanding
CO2	Identify the operation, characteristics and applications of DC motors	14	Understanding
CO3	Develop equivalent circuit of a single phase transformer	14	Applying

CO4	Apply various tests to calculate the efficiency and regulation of a single phase transformer	15	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	3						
CO4		3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the classification and characteristics of DC generators		
M1.01	Illustrate the constructional details of DC generator	3	Understanding
M1.02	Identify the classification of dc generators	5	Applying
M1.03	Explain armature reaction in dc generators	3	Understanding
M1.04	Illustrate various characteristics of dc generators	4	Understanding
Contents: DC Generator: Working principle - Construction - materials used and functions of various parts of DC generator -Armature windings- Lap and Wave windings(basic concepts and comparison only)-Emf equation .			

DC generator classification - classification based on excitation-separately excited- self excited (Shunt, Series and Compound Generators) -voltage equation – problems -Losses in dc generator -power stages- efficiency-condition for maximum efficiency(derivation needed)- problems

Armature Reaction and commutation : Basic concept of armature reaction - Concept of commutation - methods of improving Commutation(Basic idea only).

Characteristics Of D.C. Generators: Concept of Voltage Build-up - Internal and external Characteristics(series and shunt generator only)

CO2	Identify the operation, characteristics and applications of DC motors		
M2.01	Explain the working principle of DC motor and solve various motor parameters	5	Applying
M2.02	Illustrate the performance characteristics of dc motors	3	Understanding
M2.03	Explain starting and speed control methods of dc motors	4	Understanding
M2.04	Summarize the procedures for testing of DC motors	2	Understanding
	Series Test – I	1	

Contents :

Principle of operation DC Motor: Working principle -significance of back emf-classification(listing only) - voltage equation -problems -condition for maximum power(derivation needed)- Concept of torque- armature and shaft torque (definitions and equations only) - problems

Characteristics of DC motors- Electrical and Mechanical characteristics of Shunt , Series and Compound Motors- applications of motors- Power Flow Diagram-condition for maximum efficiency

Starting and Speed control of DC Motors -Starting of DC motors- Necessity of Starters- schematic diagram and working of three point and four point starters- -speed control methods - various methods- flux control and armature control (shunt and series motors only)

Testing of DC Machines : various tests - Swinburne's Test, Load Test(diagram and procedure only)

CO3	Develop equivalent circuit of a single phase transformer		
M3.01	Outline the working principle of a transformer	4	Understanding
M3.02	Derive the EMF equation of a transformer and solve problems.		Applying
M3.03	Illustrate the phasor diagrams of transformers.		Understanding
M3.04	Construct the equivalent circuit of a single phase transformer		Applying
Contents : Single phase transformers-Working principle-constructional details-classifications – core type - shell type Emf equation - derivation of emf equation-Transformation ratio - Simple problems-ideal transformer -concept-working - phasor diagram Phasor diagrams of transformer - concept of practical transformer -phasor diagrams-on no load and load – Phasor diagrams for various power factors Equivalent circuit - equivalent values of winding parameters referred to primary and secondary -equivalent circuit of a transformer-problems			
CO4	Apply various tests to calculate the efficiency and regulation of a single phase transformer.		
M4.01	Solve the efficiency and regulation of a single phase transformer	4	Applying
M4.02	Make use of OC & SC tests to predetermine the efficiency and regulation of single phase transformers.	3	Applying
M4.03	Explain the working of an autotransformer and instrument transformers	4	Understanding
M4.04	Illustrate various inter-connections of three phase transformers	4	Understanding
	Series Test – II	1	

Contents :

Efficiency and Regulation: Losses and efficiency of a transformer -derive the condition for maximum efficiency – problems-Voltage Regulation of a Transformer - concept of regulation up and down - equations - simple problems

Testing of transformers– diagram and procedures-polarity test, open circuit test, short circuit test, back to back test – rating of transformer - predetermination of efficiency and regulation-problems

Autotransformer-Working principle with basic diagram – derivation of saving of copper – advantages – disadvantages – applications.

Instrument transformers-CT-PT-Working - applications

Three phase transformer –Basic concepts-various three phase transformer connections – star-star, star-delta, delta-delta and delta-star- diagrams only .

Text / Reference

T/R	Book Title/Author
T1	P S Bimbhra, “Electrical Machines”, Khanna publications, Edition: 2, 2021
T2	J B Gupta, “Theory and Performance of Electrical Machines”, S K Kataria & Sons publications, 2013
R1	I J Nagrath, D P Kothari, “Electrical Machines”, Tata Mc Graw Hill publications, 4 th Edition,

Online Resources

Sl.No	Website Link
1	NPTEL courses: https://nptel.ac.in/courses/108/102/108102146
2	https://nptel.ac.in/courses/108/105/108105155
3	https://nptel.ac.in/courses/108/106/108106071
4	https://nptel.ac.in/courses/108/105/108105053
5	https://nptel.ac.in/courses/108/105/108105017/
6	www.swayam.gov.in
7	Virtual labs : http://vlabs.iitb.ac.in/vlab/labsec.html

8	http://em-coep.vlabs.ac.in
9	http://vem-iitg.vlabs.ac.in
10	NPTEL channel: www.youtube.com/channel/UC640y4UvDAIya_WOj5U4pfA
11	www.javatpoint.com/electrical-machines-tutorial
12	https://1filedownload.com/wp-content/uploads/2019/08/Electric-Machines-By-D.P.KothariI.J.Nagrath.pdf
13	https://www.electrical4u.com/electric-machines/